(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

VI Semester, April, 2024

Paper Code: OCSE-310T **Subject: Data Structures and Algorithms**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	What are linear and non-linear data structure, explain with examples.?	3	CO1 & CO2
(b)	Differentiate between arrays and linked lists.	2	CO1
(c)	Explain Binary search tree.	3	CO2
(d)	Explain the Row Major and Column Major Order method of array representation.	2	CO1
2 (a)	A 2-dimensional array X[4] [4] is stored row-wise in the memory. The 1st element of the array is stored at location 80. Find the memory location of X[3] [1] if each element of array required 4 byte memory.	5	CO1
(b)	Explain Priority Queues and list their applications.	5	CO2
3(a)	Consider the following infix expression and convert into postfix expression using stack. A+B*C-D^(E^F)	4	CO1
(b)	Give the algorithm/program for insertion and deletion of a node in binary search tree.	6	CO2
4(a)	What do you understand by the term Algorithm, explain the importance of space-time tradeoff ?	5	CO1 & CO2
(b)	Explain Preorder, Postorder, and Inorder Tree traversal.	5	CO2